



Evidence-Guided Multimodal Fake News Detection using Text-Image Consistency and External Knowledge

School of Computer Science and Engineering – Winter Semester 2025 - 26

Introduction

Social media has accelerated the spread of misinformation, where fake news often combines misleading text with manipulated or unrelated images. Traditional detection systems primarily rely on textual analysis and fail to verify claims using real-world evidence. This creates a gap in identifying credibility and providing explanations. To address this, a multimodal and evidence-driven approach is required for reliable fake news detection.

Motivation

The rapid spread of misinformation on social media, especially through combined text and images, makes detection challenging and unreliable. Where most existing models classify content without verifying factual correctness or providing clear explanations. This motivated the development of an evidence driven system.

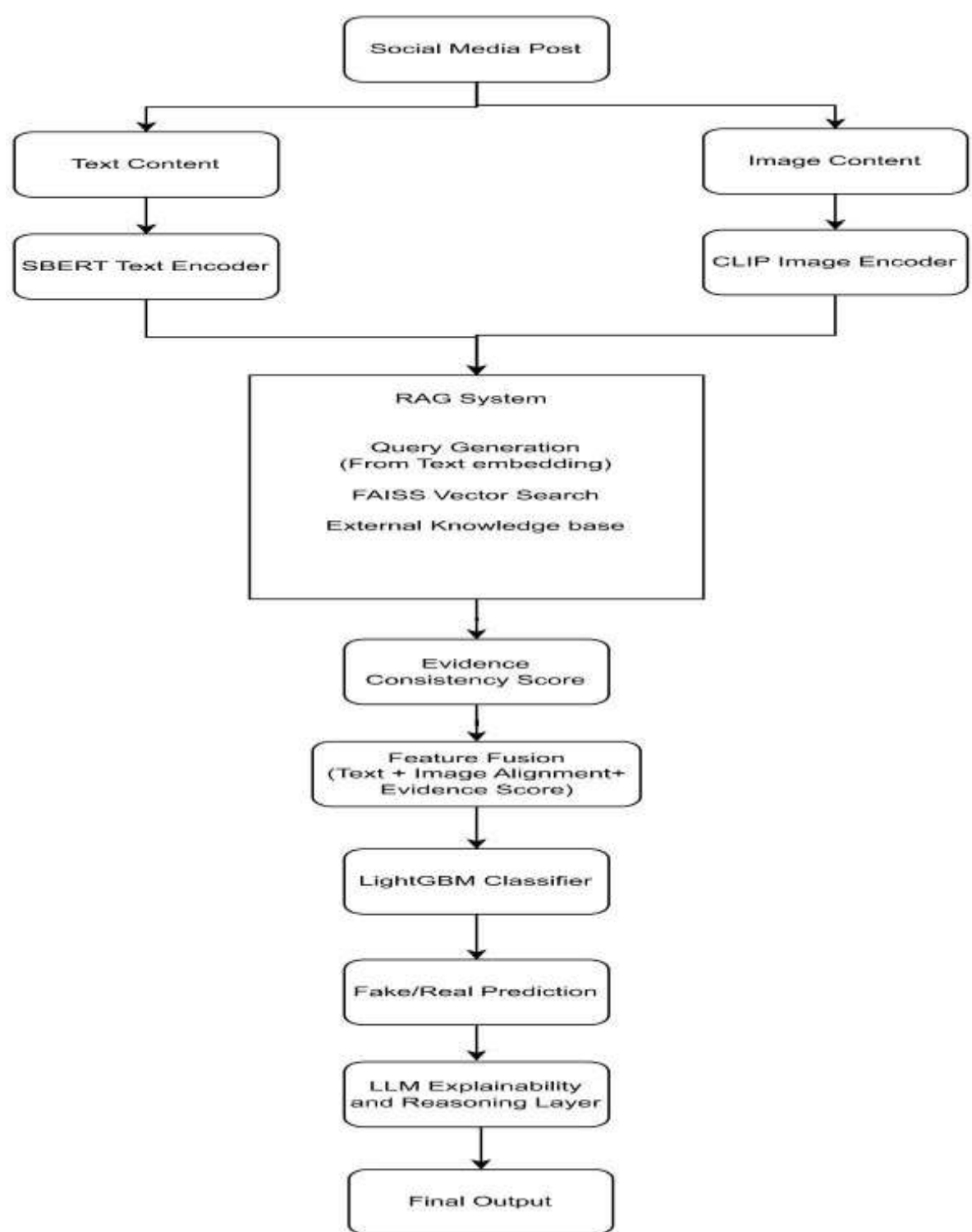
Scope of the Project

This project focuses on detecting fake news by analyzing both textual content with associated images using multimodal approach.

It incorporates external evidence retrieval to verify the credibility of claims and improve decision reliability.

The system is designed to provide not only predictions but also explanations supported by retrieved evidence and applicable to social media content and can be extended for real time and largescale misinformation detection

Methodology



This System Follows a structured multimodal pipeline that integrates textual understanding, visual verification and evidence – based verification combined with LLM Reasoning on basis of improving fake news detection reliability.

1. Textual Semantic Representation: The textual content of social media posts is processed using Sentence-BERT to generate dense semantic embeddings. This enables the system to capture contextual meaning and relationships between words, allowing accurate representation of the claim beyond simple keyword matching.

2. Image–Text Consistency Analysis: To ensure the reliability of visual information, the system uses CLIP to evaluate the alignment between textual claims and associated images. By computing similarity scores in a shared embedding space, the system identifies whether the image supports or contradicts the given text.

3. Evidence Retrieval using RAG: To validate the claim, a retrieval-based approach is employed where relevant documents are fetched from external knowledge sources. The SBERT embeddings are used as queries, and FAISS enables efficient similarity-based search to obtain supporting evidence.

4. Evidence Consistency Evaluation: The retrieved evidence is further analyzed using a Natural Language Inference model to determine whether it supports or contradicts the claim. This step produces an evidence consistency score, improving the credibility of the system’s decision.

5. Feature Fusion and Classification: All extracted features, including textual embeddings, image alignment scores, and evidence consistency, are combined into a unified representation. This is passed to a LightGBM classifier, which predicts whether the news is fake or real.

6. Explainability and Final Output: An LLM-based layer generates human-readable explanations using the retrieved evidence. The final output includes prediction, confidence, supporting evidence, and explanation, making the system both accurate and interpretable.

Results

The proposed evidence-guided multimodal framework was evaluated using the FakeReddit dataset with 20,000 samples. The system integrates textual, visual, and evidence-based features to improve classification performance.

The final prediction is obtained using the LightGBM ensemble model, expressed as:

$$\hat{y} = \sum_{m=1}^M f_m(x)$$

where $f_m(x)$ represents the prediction from each decision tree and M is the number of trees. This allows the model to learn complex patterns from multimodal features.

RAG Retrieval Similarity

The retrieved evidence module identifies the most supporting evidence by comparing claim embedding with evidence embeddings stored in FAISS index.

The similarity between vectors is computed using Euclidean distance.

$$d(q, xi) = \|q - xi\|_2$$

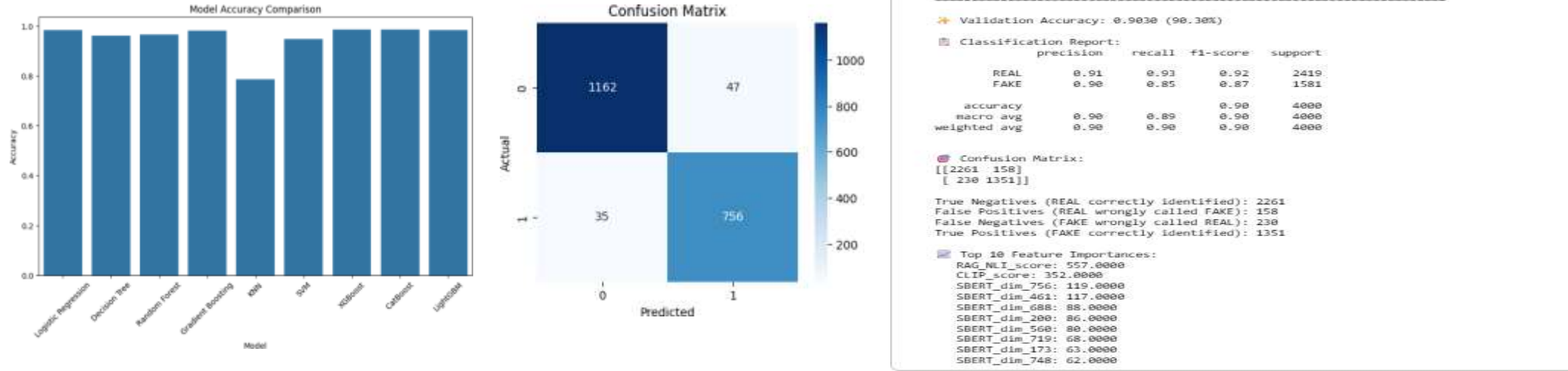
Model Comparisons and Ablation study on Performance :

Table I – Model Comparisons Results

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.9835	0.979747	0.978508	0.979127
Decision Tree	0.9605	0.939506	0.962073	0.950656
Random Forest	0.9655	0.973753	0.938053	0.955570
Gradient Boosting	0.9815	0.987080	0.965866	0.976358
KNN	0.7855	0.738158	0.709229	0.723404
SVM	0.9475	0.938619	0.927839	0.933249
XGBoost	0.9845	0.990956	0.969659	0.980192
LightGBM	0.9835	0.992208	0.965866	0.978860
LSTM	0.976	0.9856	0.9532	0.9691

configuration	Accuracy	Precision	Recall	F1
Text only	0.7970	0.739726	0.750948	0.745295
Text + CLIP	0.8020	0.749684	0.749684	0.749684
Text + RAG	0.9845	0.992228	0.968394	0.980166
Full Model	0.9835	0.990933	0.967130	0.978887

The ablation results highlight that the inclusion of RAG-based evidence retrieval contributes the most to performance improvement, while multimodal features further enhance accuracy.



The model achieved a validation accuracy of **90.30%**, demonstrating strong capability in detecting both fake and real news. These results indicate that combining multimodal features with evidence retrieval significantly improves detection reliability.

Agentic RAG Fake News Detection System

This AI system verifies news claims using:

- Retrieval: Evidence based on RAG Wikipedia database CLIP Model for Image-Text embedding
- LLM reasoning
- Image verification

User News Claim

Former President Joe Biden's cancer is a form of "turbo cancer" caused by mRNA vaccines.

(upload an image optional)

Drag and drop file here
Upload (JPG, PNG, GIF, PDF, PPT, DOCX)

Verify Claim

Prediction

Prediction: FALSE Confidence: 1.0 Explanation: The claim that "Former President Joe Biden's cancer is a form of 'turbo cancer' caused by mRNA vaccines" is false. Multiple fact-checking sources directly refute this claim.

- No evidence for "turbo cancer": The provided evidence consistently states that there is no such medical phenomenon as "turbo cancer" or "very aggressive cancer" caused by residual DNA from vaccines. (factcheck.org, postifix/facebook.com)
- Biden's cancer not linked to vaccines: Fact-checkers explicitly state that "Biden's cancer not caused by vaccines" (dailynews.com) and that "COVID-19 vaccines are not linked to 'turbo cancer'" (todaynews.com). Biden was diagnosed with prostate cancer, not a "turbo cancer" (dailynews.com).
- Moderna did not admit this: The claim that Moderna admitted mRNA vaccines cause "turbo-cancer" is also false (factcheck.org, todaynews.com).

The image, which depicts Joe Biden speaking at a podium in a formal setting, provides no information

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User News Claim

IND vs ENG: India enter Bazball Universe with concerns about efficacy of turning tracks. As the Bazball sends in India, with the first Test set to begin on Thursday in Hyderabad, all things point out to why the series could be the most challenging one for India.

(upload an image optional)

Drag and drop file here
Upload (JPG, PNG, GIF, PDF, PPT, DOCX)

Verify Claim

Prediction

Prediction: TRUE Confidence: 0.95 Explanation: The claim states that India is entering the Bazball Universe with concerns about the efficacy of turning tracks, and that the series will be challenging for India, with the first Test in Hyderabad.

The provided evidence strongly supports this claim:

- Direct Match: The first piece of evidence from "Indianexpress.com" uses almost identical phrasing: "IND vs ENG: India enter Bazball Universe with concerns about efficacy of turning tracks. As the Bazball sends in India, with the first Test set to begin on Thursday in Hyderabad, all things point out to why the series could be the most challenging one for India."
- Series Context and Challenges: The "apnews.com" evidence confirms the start of the five-match Test series in Hyderabad and highlights that "India also has its share of problems ahead of the first test," specifically mentioning Virat Kohli's withdrawal. This supports the idea that the series could be challenging for India.
- Discussion of Bazball's Impact: The YouTube evidence from "Buzborder" shows a debate on whether "India will worry at all about it (Bazball) in the aftermath of #afriac" indicating that India's potential

Conclusion

This work presents a multimodal fake news detection system that combines textual semantics, image–text consistency, and evidence verification. The integration of RAG and LightGBM improves reliability, while the explainability module enhances user understanding. The system demonstrates strong performance and practical applicability in real-world scenarios.

References

- W. Benaouda, S. Ouamour, and H. Sayoud, “Comparison of CNN model with different machine learning models for fake news detection,” in *Proc. Int. Conf. Electrical, Computer, Telecommunication and Energy Technologies (ECTETech)*, 2024, pp. 1–6.
- M. K. Çoban and G. Bakal, “NLP-driven fake news detection: A machine learning perspective,” in *Proc. Int. Congr. Human- Computer Interaction, Optimization and Robotic Applications (ICHORA)*, 2025, pp. 1–6.

Demo Video Link:



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